VC

RECOVERY HANDBOOK

VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

Photon OS Appliance Boot Recovery

Step-by-step recovery procedure for VCF 9 Photon OS appliances stuck in systemd emergency mode after disk or filesystem damage — using `fsck.repair=yes` via GRUB.

PHOTON OS 5 | FSCK | GRUB | EMERGENCY MODE | BOOT RECOVERY

VCF 9.0 VMware Cloud Foundation

VCF 9.0 Photon OS Appliance Boot Recovery — fsck.repair=yes via GRUB

Version: 1.0 | **Date:** May 13, 2026 | **Verified Cases:** VCSA, SDDC Manager, NSX Manager, VCF Operations, Fleet/LCM

> *This handbook covers a specific boot failure mode where a VCF Photon OS appliance halts in `systemd` emergency mode because one or more partitions cannot be mounted at boot. The cause is almost always partition-level damage (commonly from a power loss during write) — not a vSAN object accessibility issue, not a hardware fault, not a configuration drift. The recovery is non-destructive: appending `fsck.repair=yes` to the GRUB kernel command line lets `systemd-fsck` repair the affected filesystems on the next boot. Documented per Broadcom KB 326323 and verified end-to-end on multiple VCF 9.0.1 appliances (most recently SDDC Manager on 2026-05-13).*

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1. Executive Summary

Item	Value

Scope	VCF 9.0.x Photon OS 5 appliances (VCSA, SDDC Manager, NSX Manager, VCF Operations, Fleet/LCM, Operations Cloud Proxy)
Trigger condition	Appliance powers on, kernel loads, then `systemd` halts in *emergency mode* with `[TIME] Timed out waiting for device <UUID>` and `[DEPEND] Dependency failed for /<mountpoint>` messages on the console
Most common root cause	Filesystem damage from a hard power-off during write — partition GPT entries or filesystem superblocks no longer match what `/etc/fstab` references
Recovery type	**Non-destructive** — `fsck.repair=yes` is a kernel command-line parameter consumed by `systemd-fsck` on next boot; data is preserved
Expected runtime	5–30 minutes depending on filesystem size and damage extent. Small partitions (`/boot/efi` , `/boot`) finish in seconds; large partitions (`/storage` , `/var/log`) can take many minutes
Success rate	High when the underlying disks are healthy and damage is at the filesystem layer. Low when partition tables are entirely lost or vSAN-level corruption is present (see Section 8)
Verified end-to-end	2026-05-13 on SDDC Manager (VCF 9.0.1, 6-VMDK Photon appliance, recovered from systemd emergency mode after May 5 power outage corruption)

> **Bottom line.** If a VCF Photon appliance is stuck in `systemd` emergency mode and the underlying vSAN datastore reports all objects `healthy` (verify with `esxcli vsan debug object health summary get`), then `fsck.repair=yes` via GRUB is the correct first attempt. It is non-destructive, repeatable, and verified for VCF 9.0.1 Photon OS 5 appliances. Move to a redeploy path only after this fails.

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2. When to Use This Procedure (Symptoms)

Use this procedure when **all** of the following are true:

1. The target VM is a **VCF 9.0.x Photon OS appliance** — VCSA, SDDC Manager, NSX Manager, VCF Operations, Fleet/LCM, or Cloud Proxy.
2. The VM **boots into the kernel** but **halts at a `systemd` emergency-mode prompt** on the console.
3. The console shows one or more of these messages near the end of the boot output:

```
[TIME]   Timed out waiting for device <UUID-or-by-uuid-path>.
[DEPEND] Dependency failed for /boot/efi.
[DEPEND] Dependency failed for Local File Systems.
[DEPEND] Dependency failed for File-<UUID>.
[DEPEND] Dependency failed for /boot.
[FAILED] Failed to start Rule-based for Device Events and Files.
You are in emergency mode. After logging in, type "journalctl -xb" to view system
```

```
logs,
"systemctl reboot" to reboot, "systemctl default" or "exit" to boot into default
mode.
Give root password for maintenance (or press Control-D to continue):
```

4. The underlying **vSAN datastore is healthy** — verified by running:

```
# ESXi shell on any cluster host
esxcli vsan debug object health summary get
```

Expected: `inaccessible = 0`, `reduced-availability-* = 0`, `healthy = <total expected>`. If vSAN reports issues, the problem is below the filesystem layer and this procedure will not help.

5. The VMDK files for the affected VM **exist on the datastore** and are referenced in the `.vmx`:

```
# ESXi shell on the host running the VM (replace <vmid>)
vim-cmd vmsvc/get.config <vmid> | grep -E 'fileName|scsi[0-9]+:[0-9]+\.present'
```

Expected: every `scsi<X>:<Y>.present = "TRUE"` has a corresponding `scsi<X>:<Y>.fileName = "<name>.vmdk"`.

If any of these five conditions is false, **stop and triage that condition first** — fsck.repair=yes will not help, and may extend the outage by masking the real problem.

2.1 Symptoms that look similar but require a different procedure

Console message	Procedure
`[FAILED] Failed to mount /sysroot` (without device-timeout above it)	KB 326323 / this procedure — same fsck path, also applicable
`Welcome to GRUB!` and stays in GRUB prompt (no kernel loads)	Bootloader corruption — restore from snapshot, or use rescue ISO to reinstall GRUB. fsck won't help.
`Press any key to continue` then power off	Hardware-version mismatch or virtualHW config — check `.vmx` and VM compatibility
`Booting from Hard Disk` then nothing	Boot disk not selected, EFI variables lost — VM Settings → Boot Options → check Firmware = EFI and boot order

Boots fully, login works, but services don't start	Application-layer issue, NOT a boot/fs issue — see appliance-specific runbooks

3. Background: How Photon OS Appliances Boot

A Photon OS 5 appliance follows this boot sequence:

1. **Firmware (EFI)** loads GRUB from the EFI System Partition (`/boot/efi`)
2. **GRUB** loads the Linux kernel and initramfs from `/boot`
3. **Kernel** initializes hardware drivers and switches to the **initramfs**
4. **initramfs** mounts the **root filesystem (** `/` **)** based on the `root=PARTUUID=...` parameter in the kernel command line
5. **systemd** starts as PID 1, processes `/etc/fstab` , and mounts the remaining filesystems in dependency order
6. **Application services** (vmware-vpostgres, vmware-vmon, sddc-manager-deploy.service, NSX manager.service, etc.) start once filesystems are mounted

The failure mode this handbook addresses occurs at **step 5**: `/etc/fstab` references partitions by `UUID=...` or `PARTUUID=...` , and the partition with that UUID either no longer exists on disk or has a damaged superblock that `systemd-fsck` (configured to run on every mount per `/etc/fstab` options) refuses to mount without manual intervention.

3.1 The role of `fsck.repair=yes`

The kernel parameter `fsck.repair=yes` (alternatively written `systemd.fsck-repair=yes` on older systemd) instructs `systemd-fsck` to pass `-y` (answer yes to all repair prompts) when running `e2fsck` against ext4 filesystems and `xfs_repair -L` when running against XFS. This converts an interactive fsck (which would halt waiting for operator input) into an automatic repair.

Default Photon OS behavior is `fsck.repair=preen` — answer "yes" only to safe questions and otherwise halt. After a hard power loss, the damage often exceeds the `preen` -safe threshold, so the appliance halts. Setting `fsck.repair=yes` for one boot accepts the repair tradeoff: some corrupt files may end up in `/lost+found` , but the filesystem itself becomes mountable again.

3.2 Typical Photon appliance disk layout

Mount	Filesystem	Typical role	fsck tool
`/boot/efi`	vfat / FAT32	EFI System Partition, holds GRUB	`fsck.vfat`

`/boot`	ext4	Linux kernel + initramfs images	`e2fsck`
`/`	ext4	Root filesystem (rare on multi-disk appliances)	`e2fsck`
`/sysroot` (initramfs view of `/`)	ext4	Same as `/` after pivot	`e2fsck`
`/storage/*`	ext4	Appliance state (postgres, configs, certs, logs)	`e2fsck`
`/var/log`	ext4	systemd journal + application logs	`e2fsck`

Multi-disk appliances (SDDC Manager has 6 VMDKs, VCSA 9 has 10+ VMDKs in some sizes) spread these mountpoints across separate VMDKs; damage to any one can produce the emergency-mode console.

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4. Prerequisites

Before starting, confirm or have ready:

Prerequisite	How to verify
VM console access	vSphere Client → VM → Summary → **Launch Web Console** (or **Launch Remote Console** if VMRC is installed). Confirm you can see the systemd emergency prompt.
vSAN health (if VM is on vSAN)	`esxcli vsan debug object health summary get` on any cluster host shows `healthy = <expected>, inaccessible = 0`
VMDKs present on datastore	`vim-cmd vmsvc/get.config <vmid>` on the running host — verify all expected `scsi*.fileName` entries match files in the VM's datastore folder
Appliance root password (only needed if fsck.repair fails and you drop to maintenance shell)	From the vault — for VCSA / SDDC Manager / NSX Manager / VCF Operations these are typically operator-set at OVA deploy time
No active vSAN resync touching this VM	`esxcli vsan debug object health summary get` shows `data-move = 0` and `reduced-availability-with-active-rebuild = 0`
Power state of dependent VMs is acceptable to lose briefly	The appliance you are recovering will be **reset** (hard power-cycle equivalent). If other services depend on it, plan their state accordingly

> **Do not attempt this procedure on an appliance whose VMDKs are on a vSAN datastore showing `inaccessible` objects.** Forcing fsck against an inaccessible-from-vSAN VMDK can corrupt the on-disk filesystem further. Repair vSAN first using KB 327032 or the cluster-level Skyline Health "*vCenter state is authoritative → Remediate Inconsistent Configuration*" action.

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5. Step-by-Step Recovery Procedure

5.1 Step 1 — Open the VM console

1. In the vSphere Client, navigate to the affected appliance VM.
2. Right-click the VM → **Open Remote Console** (VMRC) or **Launch Web Console**. Confirm the systemd emergency prompt is visible.
3. **Do NOT type the root password yet.** You will instead reset and intercept GRUB.

5.2 Step 2 — Reset the VM (hard power-cycle)

1. From the vSphere Client (not from inside the guest console): right-click VM → **Power** → **Reset**.
2. **Alternative if you have already entered the maintenance shell and typed the root password:** run `systemctl reboot` at the maintenance prompt.
3. Keep the console window in focus immediately — the GRUB menu appears briefly (typically 3–5 seconds).

5.3 Step 3 — Intercept GRUB and edit the kernel command line

1. When the GRUB menu appears, press the **up/down arrow keys** to highlight the **default Photon kernel entry** (typically the top entry, labeled `Photon Linux <version>`).
2. Press `e` (lowercase) to enter the edit screen. **Do NOT press Enter** — that boots the entry unmodified.
3. In the edit screen, use the arrow keys to find the line that begins with `linux` — this is the kernel command line. The line looks like:

```
linux /boot/vmlinuz-<version>-photon-<release> root=PARTUUID=<uuid> ro <other-params>
```

4. Use the End key (or arrow keys) to position the cursor at the **end of that `linux` line**.
5. **Append a single space and then `fsck.repair=yes`:**

```
linux /boot/vmlinuz-... root=PARTUUID=... ro ... fsck.repair=yes
```

6. **Do NOT alter anything else** on that line — do not modify the `root=` parameter, do not remove `ro`.

5.4 Step 4 — Boot with the modified command line

1. Press `Ctrl-X` (or **F10**) to boot.
2. The kernel loads and initramfs starts. After a few seconds you should see `systemd-fsck` output beginning to scroll past on the console (see Section 6 for what to expect).

The edit is **transient** — it applies only to this boot. After a successful repair, subsequent boots use the unmodified GRUB entry and run `fsck.repair=preen` again (which will not find further damage on the now-repaired filesystems).

5.5 Step 5 — Wait for fsck to finish on each filesystem

1. fsck runs sequentially against each `/etc/fstab` entry that has a non-zero `passno` (sixth column). The order is:
   - `/` (passno 1) — runs first, typically already mounted read-only by initramfs by this point
   - `/boot`, `/boot/efi`, `/storage/*`, `/var/log` (passno 2) — run in parallel by default, but practically one device at a time on a single virtual disk
2. **Do not interrupt.** Some filesystems can take 5–30 minutes if heavily damaged.
3. Once all filesystems are clean, `systemd` continues the boot:
   - Mounts the filesystems read-write
   - Starts the appliance services (postgres, web UIs, etc.)
   - Reaches the login prompt (or the appliance's specific welcome banner)

5.6 Step 6 — Verify recovery

After the appliance reaches the login prompt:

1. Log in as `root` with the appliance's root password.
2. Verify all expected filesystems are mounted:

```
# appliance shell
df -h
```

Compare against your appliance's normal mount layout (VCSA, SDDC Manager, etc. have appliance-specific layouts). All expected mounts should appear.

3. Look for journal errors from the most recent boot:

```
# appliance shell
journalctl -b -p err --no-pager | head -100
```

You will see entries about `lost+found` and the recent fsck pass; these are expected. Investigate any service-level errors (postgres start failure, etc.) separately.

4. Verify appliance-specific health:

Appliance	Health command
VCSA	`service-control --status --all` then check `https://<fqdn>/ui`
SDDC Manager	`systemctl status sddc-manager` and `curl -sk https://localhost/v1/system`
NSX Manager	`get cluster status` in `nsxcli`, or `https://<fqdn>` UI
VCF Operations	`systemctl status vmware-vcops` and login at `https://<fqdn>`
Fleet/LCM	`systemctl status vrlcm-server` and `https://<fqdn>` UI

5. **If services started cleanly**, recovery is complete — the appliance is back in operation.

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6. Expected Output During fsck

6.1 Healthy progression

A typical successful fsck pass on a Photon appliance prints output like the following to the console (one block per filesystem):

```
[  X.XXX] systemd-fsck[NNN]: /dev/sdaN: clean, INODES/INODES files, BLOCKS/BLOCKS
blocks
```

...for filesystems that were already clean (initial check finds no damage).

For filesystems being repaired, output is more detailed:

```
[  X.XXX] systemd-fsck[NNN]: Pass 1: Checking inodes, blocks, and sizes
[  X.XXX] systemd-fsck[NNN]: Pass 2: Checking directory structure
[  X.XXX] systemd-fsck[NNN]: Pass 3: Checking directory connectivity
[  X.XXX] systemd-fsck[NNN]: Pass 4: Checking reference counts
[  X.XXX] systemd-fsck[NNN]: Pass 5: Checking group summary information
[  X.XXX] systemd-fsck[NNN]: /dev/sdaN: ***** FILE SYSTEM WAS MODIFIED *****
[  X.XXX] systemd-fsck[NNN]: /dev/sdaN: <NN>/<NN> files, <NNN>/<NNN> blocks
```

The line `***** FILE SYSTEM WAS MODIFIED *****` indicates fsck applied repairs successfully.

6.2 Output blocks that are normal (do not panic)

- `Inode <N>, i_blocks is <X>, should be <Y>. Fix? yes` — fsck found and fixed an inconsistent inode block count. `fsck.repair=yes` answers automatically.

- `Reconnect to /lost+found? yes` — fsck found an orphaned inode (file disappeared from its directory due to incomplete writes) and reconnected it to `/lost+found`. Files in `/lost+found` may be recoverable, but for VCF appliances they are typically transient log files or temp files that can be safely ignored.
- `Free inodes count wrong (<X>, counted=<Y>). Fix? yes` — superblock counter mismatch, fixed automatically.
- `/dev/sdaN: UNEXPECTED INCONSISTENCY; RUN fsck MANUALLY.` at the END followed by successful repair lines — this message appears once at the start of each repaired FS as the "why we're running"; the subsequent passes do the work.

6.3 Output blocks that indicate the procedure is not sufficient

- `Superblock has an invalid journal (inode <N>). Clear<y>?` followed by `***** FILE SYSTEM WAS MODIFIED *****` is OK — journal cleared, filesystem still usable.
- `Bad block in superblock` repeated across passes, or `Could not allocate <N> blocks in inode table` — physical-layer damage. Skip to Section 8.
- **fsck silently halts mid-pass and returns to systemd emergency mode without** `***** FILE SYSTEM WAS MODIFIED *****` — fsck gave up on a filesystem it cannot repair. Skip to Section 8.

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7. Success Indicators

Recovery is successful when **all** the following are true:

Check	Where to verify	What to look for
Boot reached login prompt	VM console	`<hostname> login:` (or appliance-specific banner)
`df -h` lists all expected filesystems	Appliance shell	No mountpoints missing vs. the appliance's documented layout
`journalctl -b -p err` is clean (or only shows fsck-related lines)	Appliance shell	No service-start failures, no remount-as-read-only errors
Appliance-specific service status is green	See Section 5.6 table	Web UI reachable, API responsive
`mount	grep -v ' ro,'` shows no read-only mounts of normally-writable filesystems	Appliance shell
Last fsck timestamp updated	`tune2fs -l /dev/sdaN	grep "Last checked"`

Record the recovery in the appliance's runbook / incident log:

```
2026-MM-DD HH:MM <appliance-name> recovered from systemd emergency mode via GRUB
fsck.repair=yes.
Root cause: <power outage / hard power-off / vSAN MM mishandling / etc.>
Filesystems repaired: <list from console>
Data integrity check: <results of post-recovery application-level verification>
```

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8. Troubleshooting: When fsck Doesn't Fix It

8.1 fsck completes but appliance still drops to emergency mode

Likely cause: fsck repaired the underlying ext4/vfat filesystem, but a different mount in `/etc/fstab` is still failing — e.g., an NFS mount, a tmpfs with bad options, or a `/storage` subdirectory bind-mount.

Action:

1. At the emergency prompt, enter the root password to drop to a maintenance shell.
2. View the last boot's journal: `journalctl -xb --no-pager | less`
3. Search for `Failed to mount` to find the specific mount that's failing.
4. Inspect `/etc/fstab` and comment out the failing line (prefix with `#`):

```
# appliance shell
vi /etc/fstab
# locate failing line, prefix with #
:wq
```

5. Reboot: `systemctl reboot`. The appliance should now boot, but with the commented mount inactive — investigate that mount separately once you have a working appliance.

8.2 fsck halts on a filesystem (no "FILE SYSTEM WAS MODIFIED")

Likely cause: Damage exceeds what `e2fsck -y` can repair — typically a damaged inode table or superblock chain.

Action — try backup superblock recovery:

1. Boot the VM from a **Photon OS rescue ISO** (or any Linux live ISO with `e2fsprogs`) attached as CD-ROM. (Photon OS rescue ISOs ship with most VMware appliance ISOs.)
2. Identify the damaged partition from the systemd emergency console output — e.g., `/dev/sdaN`.

3. List available backup superblocks:

```
# rescue shell
mke2fs -n /dev/sdaN     # NB: -n is *no-op*; prints superblock locations without writing
```

4. Run fsck with an alternate superblock (typically backup is at block 32768 or 98304):

```
fsck.ext4 -b 32768 -y /dev/sdaN
```

5. If the alternate superblock is also damaged, the appliance is not recoverable with fsck. Skip to Section 9.

8.3 fsck loops on the same filesystem each boot

Likely cause: The filesystem is being marked dirty on every shutdown because `/etc/fstab` has `errors=remount-ro` and the appliance is hitting a write error during shutdown.

Action:

1. After the appliance is up, check kernel ring for disk errors:

```
dmesg -T | grep -iE "error|fail|i/o" | tail -30
```

2. If repeated I/O errors, the underlying VMDK may have damage at the VMFS/vSAN layer. Verify with `vmkfstools --diskformat` on the host. If VMDK is damaged, this procedure won't help long-term — plan a redeploy.

8.4 Cannot intercept GRUB (boot is too fast)

Likely cause: GRUB timeout is set to 0 (silent boot) — happens on some appliance images.

Action — temporary edit of GRUB timeout:

1. Reset the VM and tap **Shift** or **Esc** *repeatedly* during the EFI handoff window — this forces GRUB to display the menu on most builds.
2. Alternative: from a working sibling appliance, mount the affected VM's `/boot` partition (via rescue ISO) and edit `/boot/grub2/grub.cfg` to set `set timeout=5`. Reboot.

3. Alternative: edit `/etc/default/grub` (after recovery) to set `GRUB_TIMEOUT=5` and run `grub2-mkconfig -o /boot/grub2/grub.cfg` so future boots show the menu.

8.5 Console shows GRUB error before the menu

Likely cause: `/boot/efi` or `/boot` partition is so damaged that GRUB itself can't load.

Action:

1. Reinstalling GRUB from a rescue boot is theoretically possible but rarely worth the effort for a VCF appliance — proceed directly to redeploy (Section 9).

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9. Alternative Recovery Paths

If Section 5 + Section 8 troubleshooting do not restore the appliance, escalate to one of:

9.1 Restore from backup

Appliance	Backup mechanism
VCSA	File-based backup (`https://<vcsa>:5480` → Backup) or vSphere snapshot
SDDC Manager	`https://<sddc>/ui/sddc-manager-ui-app/backup-restore` or snapshot
NSX Manager	`https://<nsx>` → System → Backup & Restore (SFTP target)
VCF Operations	`https://<vcops>/ui` → Administration → Support → Backup
Fleet/LCM	Snapshot (file-based backup limited in 9.0.x)

Caveat: restore requires a backup taken *before* the corruption event. If the corruption happened today and the last backup was last week, you lose a week of changes — which for SDDC Manager / VCF Ops may include configuration of workload domains.

9.2 Redeploy from OVA

Match the source OVA build to the rest of the environment exactly. Steps depend on appliance:

Appliance	Deploy method	Reference
VCSA	Host Client OVA upload (NOT VCSA UI Installer if target is vSAN-only)	[[feedback_vcsa_use_host_client_ova]] + [[reference_vcsa_redeploy_ovf_properties]]
SDDC Manager	Standalone OVA via `ovftool` against vCenter	RCA §13.4 line 891

NSX Manager	PowerCLI `Import-VApp` (NOT ovftool 5.0 — vAppConfig drop bug)	[[reference_powercli_import_vapp_for_nsx9]]
VCF Operations	Standalone OVA via `ovftool`	RCA §13.4
Fleet/LCM	Standalone OVA via `ovftool`	RCA §13.4

After redeploy, the new appliance has a fresh empty database. Reconfiguration (workload domain re-import, NSX TNP recreation, cert chain re-trust, etc.) follows appliance-specific runbooks.

9.3 Snapshot rollback

If a vSphere snapshot was taken before the corruption event (often automatic before VCF upgrade operations), revert to it:

1. vSphere Client → VM → Snapshots → **Revert to Snapshot**
2. Power on the VM
3. Verify it boots normally

Snapshots older than a few days may have their own staleness issues for VCF appliances (DB drift vs. peer appliances) — verify after revert.

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10. Related KBs, Memories, and Cross-References

10.1 Broadcom KBs

KB	Title	Relevance
KB 326323	VCF appliance stuck on `/sysroot` mount fail — `fsck.repair=yes` recovery	Direct parent KB for this procedure
KB 327032	Add host back to vSAN after rebuild — verify vSAN healthy before fsck	Section 4 prerequisite check
KB 326587	Migrating vSAN stretch cluster to redeployed vCenter	Adjacent — when fsck is not the right path

10.2 Memory references

Memory	Why relevant
[[reference_vcf_photon_fsck_recovery]]	Pointer memory to this exact procedure (KB 326323)
[[feedback_vsan_resync_first_check]]	Always check vSAN before powering on / recovering mgmt VMs

[[feedback_homelab_extreme_caution]]	Read user's own runbooks first before risky actions
[[project_power_outage_20260505]]	Common trigger event for this kind of corruption
[[feedback_no_hibernation_with_vmware]]	Prevent the most common host-side cause of guest corruption

10.3 Local runbook references

Document	Section
`VCF9-Management-Plane-Loss-RCA-2026-05-01.md`	§7.3 (Tier 3 redeploy decision tree); §13.4 (standalone OVA deploy procedures)
`VCF9-Host-vCenter-Rebuild-Plan.md`	§6.2 (vCenter recovery path)
`VCF-Undocumented-Issues-Reference.md`	Issue #41 (`vAppConfig=null` is misleading); Issue #42 (VCSA UI Installer + vSAN)
`VCF_Lab_Shutdown_Startup_Guide.md`	Graceful shutdown procedure to *prevent* the corruption this handbook recovers from

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11. Document Information

Field	Value
Document Title	VCF 9.0 Photon OS Appliance Boot Recovery — fsck.repair=yes via GRUB
Version	1.0
Author	Virtual Control LLC
Date Created	May 13, 2026
Last Updated	May 13, 2026
Environment	VMware Cloud Foundation 9.0.1 / Nested Lab (4-host vSAN OSA)
Verified Cases	SDDC Manager 2026-05-13 (Photon OS 5.0, 6-VMDK, recovered after May 5 power outage corruption); pattern applies to VCSA, NSX Manager, VCF Operations, Fleet/LCM
Audience	VCF operators recovering Photon OS appliances stuck in `systemd` emergency mode
Prerequisites	VM console access (Host Client or vSphere Client), familiarity with GRUB edit screen, appliance root password
Risk Profile	Non-destructive when prerequisites in Section 4 are met. Destructive paths (Section 9) are clearly labeled

Companion Documents	`VCF9-Management-Plane-Loss-RCA-2026-05-01`, `VCF9-Host-vCenter-Rebuild-Plan`, `VCF-Undocumented-Issues-Reference`

Revision History

Version	Date	Change
1.0	May 13, 2026	Initial release. Procedure verified end-to-end on SDDC Manager 2026-05-13 after May 5 power outage corruption. Sections 5 (procedure) and 6 (expected output) reflect direct observation of that recovery.

This document is part of the VCF 9.0 Lab Documentation Suite:

- **VCF9-Master-Bible.pdf** — Complete VCF 9.0 reference guide
- **VCF_Troubleshooting_Handbook.pdf** — General troubleshooting procedures
- **VCF9-Management-Plane-Loss-RCA-2026-05-01.pdf** — Source incident that established the recovery patterns
- **VCF-Undocumented-Issues-Reference.pdf** — Catalog of undocumented edge cases
- **VCF_Lab_Shutdown_Startup_Guide.pdf** — Prevention via graceful shutdown